

ResQGrid AI

Intelligent Emergency Resource Network

AI recommends. Humans approve.

Every important decision is explainable and auditable.

IntelliCon '26 | Gate 3 Final Submission

Deshan Lakshitha | October 2026

Live: <https://web-two-mu-edisuy5tdl.vercel.app>

Repo: <https://github.com/deshanlakshitha/resqgrid-ai-v1>

The Problem

330+ deaths. 300,000+ displaced. Every monsoon.

Sri Lanka faces severe flooding almost every year (World Bank: 40 consecutive years of damage).

The DMC duty officer must answer 6 questions — simultaneously, in real time:

1. Which incidents are most urgent?
2. Who is at risk?
3. What resources are available right now?
4. Which resource goes where?
5. Which routes are passable?
6. What changed since my last decision?

Today's tools: phone calls, WhatsApp voice notes, paper logs.

Result: misdirected resources, delayed triage, undocumented decisions.

Named User

The DMC Duty Officer

Role: Runs district-level disaster response from a single operations room.

Pain: Unstructured, competing reports with no automated prioritization.

Need: A system that reads, ranks, and recommends — while keeping the human in control.

Secondary users:

- Field Responder — receives clear, approved task on mobile
- Citizen — plain-language incident reporting (no app install)
- Coordinator — audit trail of every decision

Slide 4 / 12

The Solution

ResQGrid AI — Full-Stack Emergency Decision Support

Citizen / Operator reports incident in plain language

↓

AI Triage (Alibaba Qwen) — structures report, extracts severity

↓

Priority Engine — deterministic, explainable 6-factor score

↓

Resource Matcher — optimal unit by type, proximity, hazard, route

↓

Human Approval — dispatcher approves every dispatch (enforced in code)

↓

Responder Lifecycle — accepts, travels, arrives, completes

↓

Re-plan — blocked road? System proposes alternative automatically

Slide 5 / 12

AI is Load-Bearing

Remove AI → quality degrades. Remove AI → system keeps running.

AI Triage (Qwen): free-text → typed incident schema in <2 seconds

Ensemble design: Local lexicon (40%) + Qwen LLM (60%)

Schema validation: LLM cannot write arbitrary data to the database

Graceful degradation: falls back to rule-based triage if Qwen is offline

Priority Score: 100% deterministic — 6 weighted factors, fully decomposable

Life risk (30%) | Medical urgency (25%) | People at risk (20%)

Hazard proximity (10%) | Time elapsed (10%) | Evidence confidence (5%)

Command Assistant: natural-language Q&A; from live database context via Qwen

Image Analysis: multimodal evidence assessment (advisory only)

Human-in-the-Loop by Design

AI recommends. Humans approve. Audit logs everything.

- ✓ AI never dispatches a resource without dispatcher approval
- ✓ Approval is a required API call — not just a UI checkbox
- ✓ Every AI output and human decision is written to the audit log
- ✓ Priority scores decompose into factors — no black box
- ✓ Triage results show confidence and source (lexicon vs LLM)

RBAC — 4 roles enforced server-side:

Citizen → report only

Responder → accept and advance own assignments

Dispatcher → full operations + approval

Admin → full access + audit read

Technology Stack

Production-grade. Open-source. Deployed today.

Frontend: Next.js 14 / TypeScript / Tailwind CSS / MapLibre GL JS

Backend: Python 3.11 / FastAPI / SQLAlchemy 2.0 / Alembic

Database: PostgreSQL 16 + PostGIS 3.4 (geospatial queries)

Cache/Queue: Redis 7

AI: Alibaba Cloud Model Studio — Qwen-plus

Storage: Alibaba Cloud OSS (evidence images)

Auth: JWT + RBAC (4 roles, enforced server-side)

Mobile: PWA + Capacitor Android APK

Deploy: Docker Compose / Vercel / Render

Optimizer: Hungarian algorithm (global resource allocation)

Live Demo

Not a mockup — real deployment, real database

Frontend: <https://web-two-mu-edisuy5tdl.vercel.app>

API docs: <https://resqgrid-api-9wns.onrender.com/api/v1/docs>

Demo credentials:

Dispatcher: `dispatcher@resqgrid.local` / `dispatch123`

Responder: `responder1@resqgrid.local` / `responder123`

Seeded demo data:

20 fictional incidents | 30 resources | 5 shelters

3 hospitals | 5 hazard zones | 3 blocked roads

Demo flow: Citizen reports → AI triages → Priority scored →

Resource matched → Dispatcher approves → Responder accepts →

Road blocked → System re-plans

Slide 9 / 12

What Was Built

40 commits. 10 build phases. Fully functional.

Phase 1: Architecture, DB schema, environment setup

Phase 2: Auth/RBAC, incidents, resources, audit logs

Phase 3: Dashboard UI, interactive MapLibre map

Phase 4: Deterministic priority scoring engine

Phase 5: Alibaba Qwen AI triage adapter + ensemble

Phase 6: Resource recommendations + human approval flow

Phase 7: Hazard zones, route risk, dynamic re-planning

Phase 8: Evidence/image upload pipeline (Alibaba OSS)

Phase 9: Offline-first incident outbox (IndexedDB)

Phase 10: Security hardening, LLM resilience, demo polish

Slide 10 / 12

Business Case

Feasible. Affordable. Ready for a pilot today.

Current deployment cost: ~USD 0/month (free-tier cloud)

One-command setup: `docker compose up`

No proprietary lock-in: open-source stack

Go-to-market path:

Month 1–3: Pilot at one DMC district office (fictional data)

Month 3–9: Sinhala/Tamil intake + NDRSC data feed integration

Month 9–18: All 25 districts + IoT flood sensor triggers

Year 2+: Government SaaS licence or humanitarian grant funding

Funding model: Government licence | UNDP/World Bank grants | NGO partnerships

Slide 11 / 12

Competitive Differentiation

The first AI-native, human-supervised, open-source emergency dispatch prototype for Sri Lanka.

Sahana Eden: Open source, no AI, no approval loop

WebEOC: Proprietary, no AI, no Sri Lanka fit

Ushahidi: Crowd reporting, no triage, no dispatch

ResQGrid AI: AI triage ✓ | Human approval ✓ | Open source ✓ | Sri Lanka ✓

Key differentiators:

- Explainable AI — every score is decomposable
- Human-in-the-loop enforced at the API level
- Designed for the DMC duty officer workflow
- Works offline; syncs on reconnect
- Installs as PWA or native Android APK

Slide 12 / 12

Summary

Detect → Understand → Prioritize → Allocate → Approve → Respond → Re-plan

ResQGrid AI turns fragmented emergency reports into actionable decisions.

- ✓ AI does the reading — humans make the call
- ✓ Every score is explainable — no black box
- ✓ Every dispatch requires human approval — enforced in code
- ✓ System degrades gracefully — never fails completely
- ✓ Live today — not a concept, not a mockup

In an emergency, seconds matter.

ResQGrid AI saves seconds — and keeps humans in control.

GitHub: <https://github.com/deshanlakshitha/resqgrid-ai-v1>

Live app: <https://web-two-mu-edisuy5tdl.vercel.app>
